

GELU#

[https://pytorch.org/docs/stable/generated/torch.nn.modules.activation.GELU.h](https://pytorch.org/docs/stable/generated/torch.nn.modules.activation.GELU.html)

Description:

Applies the Gaussian Error Linear Units function. where $\Phi(x)$ is the Cumulative Distribution Function for Gaussian Distribution. When the approximate argument is 'tanh', Gelu is estimated with: approximate (str, optional) – the gelu approximation algorithm to use: 'none' | 'tanh'. Default: 'none' Input: $(*)^{(*)}(*)$, where $*^{**}$ means any number of dimensions.

Function Signature

```
class torch.nn.modules.activation.GELU(approximate='none')  
[source]#
```

Parameters

Shape:

```
extra_repr()[source]#
```

Return type

Returns:

Tensor

Code Examples

```
>>> m = nn.GELU()  
>>> input = torch.randn(2)  
>>> output = m(input)
```

Detailed Documentation

Documentation

Applies the Gaussian Error Linear Units function.

where $\Phi(x)$ is the Cumulative Distribution Function for Gaussian Distribution.

When the approximate argument is 'tanh', Gelu is estimated with:

approximate (str, optional) – the gelu approximation algorithm to use: 'none' | 'tanh'.

Default: 'none'

Input: $(*)^{(*)}^{(*)}$, where $***$ means any number of dimensions.

Output: $(*)^{(*)}^{(*)}$, same shape as the input.

Return the extra representation of the module.

Runs the forward pass.